

	Architecture Dossier			Ref: ref ADNS X00D23020993	
	CONFIGURATION AIRLINE COCKPIT			Issue:	Date:
	CMIS	1.0	5BAD	1	18-Aug-2023

CONFIGURATION AIRLINE COCKPIT

Architecture Dossier

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Date	Issue	Author	Comment
25-July-2023	1	Vijay Kumar	Version for MVP v1

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1. Introduction

1.1. Purpose of the document

The Architecture Dossier (ARD) presents the general and detailed architecture of the target solution for the CMi-S (*In-Service Configuration Management Improvement*) product. This document is initiated at the start of the product development as described in the RACI table in the manuals M20543 for Scaled Agile. At this point, the different architecture candidates are described and compared. The Architecture Dossier is completed before the Technical Release of the product/application. The solution to be produced is described in a complementary manner by two dossiers, the Architecture Dossier and the Definition and Development dossier.

- The Architecture Dossier (ARD) formalizes the architecture description of the target solution. It includes the System Requirements that prescribe the architecture at the system level.
- The Definition & Development Dossier (DDD) refines the System Requirements into Components Requirements for a specific subsystem or set of subsystems

Correct traceability must exist between both these documents. The Architecture Dossier is logically drawn up before the DDD. In particular, the DDD must attempt to describe the components and the interfaces identified in the ARD. The intent of this document is to be submitted for approval.

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1.2. Scope of the document

This Architecture Dossier documents the detailed specification for the Configuration Airline Cockpit

1.3. Terminology

#	Terminology	Description
1	AWS	Amazon Web Services
2	PCP	Airbus Public Cloud Platform
3	SAS	Skywise Authorization Services
4	ALS	Airworthiness Limitation Section
5	GSE	Ground Support Equipment
6	WCU	Write Capacity Unit
7	RCU	Read Capacity Unit
8	ACM	Aircraft Configuration Management
9	SB	Service Bulletin
10	IAM	Identity and Access Management
11	COTS	Commercial Off the Shelf Software
12	ACR	Aircraft Configuration Report
13	MIS	Maintenance Information System
14	PNR/PN	Part Number
15	MOD	Modification
16	SB	Service Bulletin
17	IPC	Illustrated Parts Catalog
18	SBR	Service Business Representative
19	DLI	Data Loading Interface
20	FTS	File Transfer Service
21	ISCOM	In-Service Configuration Management

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2. Architecture Description

2.1. Business Story

CMi's main objective is to retrieve in-service configuration from the A/C+Airline and re-distribute it.

This in-service configuration will be enabled by a Configuration Cockpit that incentivizes the retrofit of reporting equipment by providing an interface for Configuration Checking against the Allowable configuration, and allow to manage the reported results and enrich the in-service configuration before distributing it.

Improve the in-service configuration management of flying aircraft (in mod/SB and in parts), to decrease the number of configuration mismatches, improve efficiency of many Customer Service Business Lines, and enable future services and/or business models, in a long-lasting way.

The digital solutions will provide the following capabilities:

1. *Browse configuration in Part - Basics*
 - 1.1. browse conf in parts based on A/L configuration datasources ACR or MIS with basic filters like operators , ac series. and export (csv,pdf,xls,xml,acm)
 - 1.2. browse conf in SB visualise the SB configuration reported to Airbus during the aircraft in-service life,SB applicability with the associated MOD corresponding to operators, fleet and msn.
2. Checker configuration in Part - Basics
 - 2.1. visualise a fleet view, of PNR discrepancies per MSN, between ACR and As-allowed OR MIS configuration and As-allowed based on data source comparison selection with color - code indicators.
 - 2.2. A detailed FIN view with PNR with color-coded discrepancies per MSN for the whole fleet, between ACR and As-allowed OR MIS configuration and As-allowed, with possibility to filter by FIN,A/C and associated attributes.
3. ACM report Export
 - 3.1. Dedicated feature for ACM reports exports for a specific airline ,program MSNs

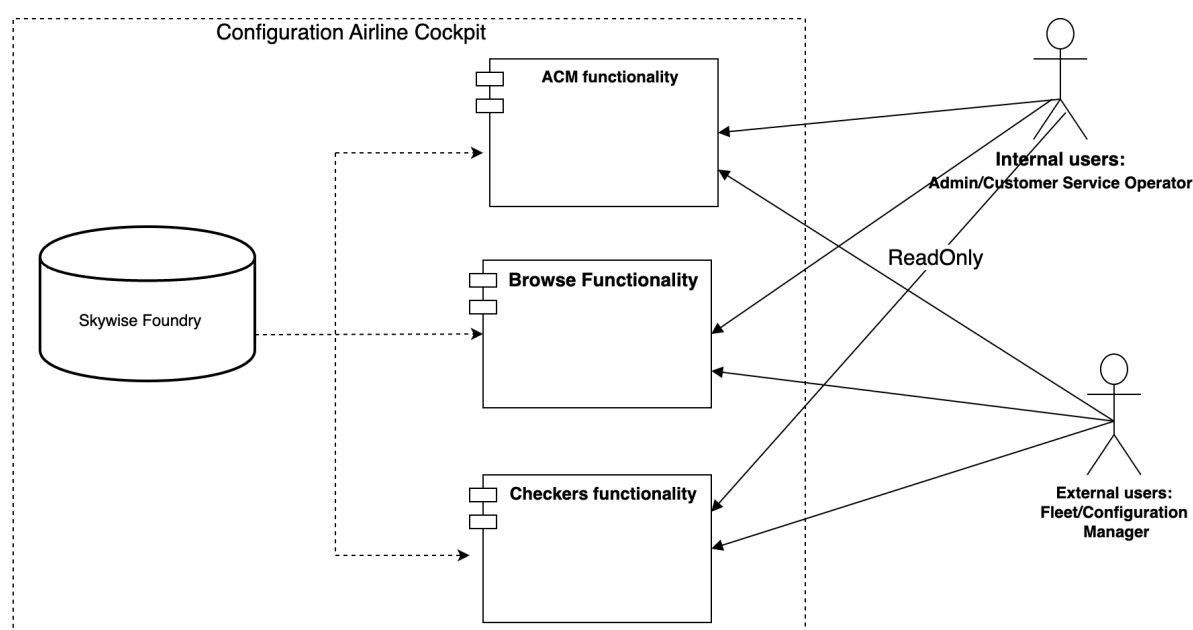
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2.2 Functional View

2.2.1. System in Context

The purpose here is to show the new system in a global context. First objective is to identify the functional boundaries of the system in this environment and how the system interacts with external actors (applications or users). Second objective is to identify how this environment is impacted by the introduction of the new system and particularly, how this environment is globally simplified (less applications, fewer interfaces). The system is seen as a black box.

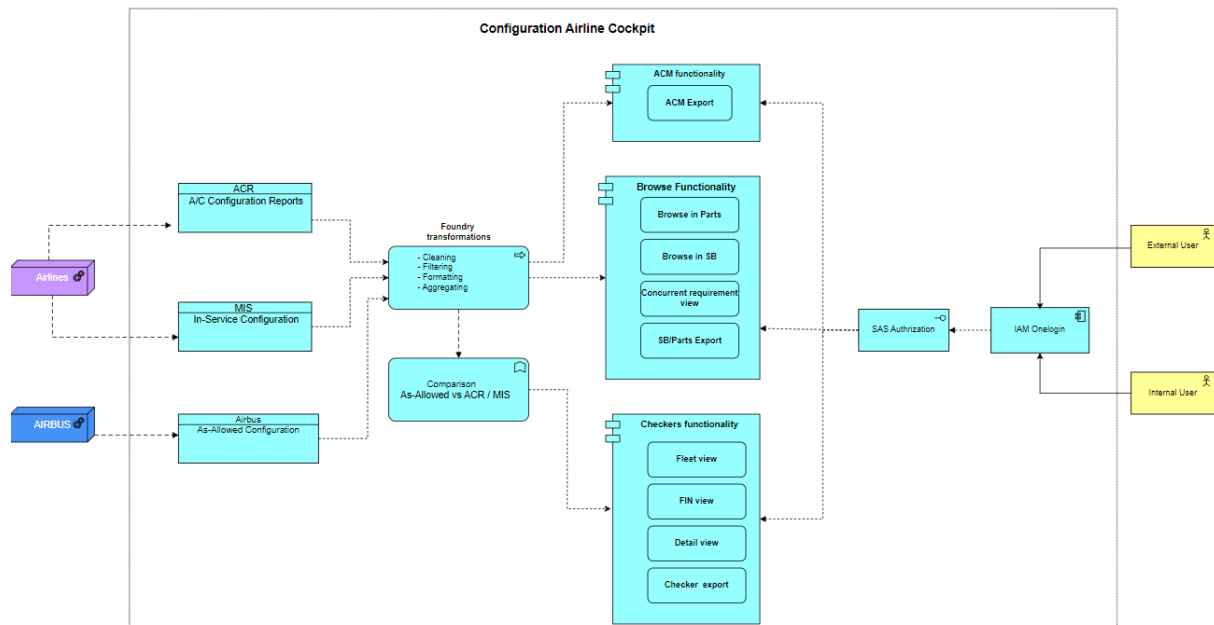
The main users of the CMi-S are from Airbus internal and Airlines(external) as well . The users can access the product as a Fleet Manager/Admin/ configuration manager OR Customer Service Operator in order to Configuration Gap analysis: In-Service vs As-Allowed to make sure the AC configuration is licit to reduce the risk of conf mismatch.



2.2.2. Functional Top Level Decomposition

The functional view is a system-level architecture that partitions the system functionalities and data into logical components and their relationships. The In-Service Configuration in parts (PNR) and in changes (SB) is built from data coming from the A/L through A/C configuration reports (ACR) and Maintenance Information Systems (MIS). The As-Allowed Configuration in parts and in changes is built from Airbus configuration management tools.

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2.2.3. Users/Audience

Below are type of users and their associated privileges

Below are application front end use cases and user permissions associated with user type:

UserType	Role Name	Browse Functionality	ACM functionality	Checkers functionality
External	Fleet Manager/Configuration Manager	Yes	Yes	Yes
Internal	Customer Service Operator	Yes	Yes	No

Where :

Yes - Read Only permission & export permission,

No - Do not have any permission

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2.3. Design Constraints

The System will be developed on top of a PaaS based on Skywise Foundry and PCP services provided by Airbus.

2.3.1. Responsibility Model

The system is hosted in two platforms, Foundry and Airbus PCP. The foundry PaaS service provides tools to extract, transform and load data.

Layer	System in Foundry	System in Airbus PCP
Data	Under Project Responsibility Skywise dataset shared responsibility between Allegiant and Skywise platform	Under Project Responsibility
Application	Under Project Responsibility	Under Project Responsibility
Runtime	Under Provider Responsibility (Palantir Foundry)	Under Provider Responsibility (AWS PCP) WebServicesProtection (Imperva) Cloud IDP (IAM)
Middleware		
OS		
Virtualization		
Servers		
Network & Storage		
Data Center		

Foundry:

Palantir	Customer
Foundry Services	<i>Customer Data</i>
Encryption in transit and at rest	<i>Customer-built Applications</i>
Network Traffic Protections	Identity and Access Management (IAM)
Operating System, Network, and Firewall Configurations	Resource Permissions (RBAC)
Infrastructure-level Monitoring and Alerting	Application-level Monitoring and Alerting
Continuous Delivery, Automated Upgrades, and Patching	
Foundation Services: → Compute → Storage → Database → Networking	
Global Infrastructure: → Regions → Availability Zones	

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Source: <https://core.skywise.com/docs/foundry/security/shared-security-responsibility-model/>

2.3.2. Standards and Security Compliance

The project shall reuse standard components as defined in Enterprise Architecture Repository (Alfabet/MyPortfolio). Adherence to Airbus Standards or Function standards is mandatory.

- The System and System's development shall rely on standard security means provided by Airbus:
 - Foundry
 - PCP ([PCP Confluence](#))
 - Airbus SecDevOps (GitHub, Jenkins, Checkmarx...)
- In addition, the System shall rely on the following standard security elements:

Name	Description	Remarks
Allowed ports and protocols	The application uses only HTTPS (default PORT 443) for the interactions with external systems	
Multi Region	The application resources are deployed only in PCP allowed AWS regions	
Internet publishing	The application uses CDK based Web Internet Publishing pattern published in PCP blueprints	
TLS & Ciphers Web Access	Security policy TLSv1.2_2021 Certificate Signature Algorithm PKCS #1 SHA-256 With RSA Encryption	
Firewall	Imperva as a firewall and content distribution network (CDN)	
Internet publishing	The application uses CDK based Web Internet Publishing pattern published in PCP blueprints	
authentication	OneLogin for user authentication	

2.3.3. Data Protection and Compliance

This following section describes the data classification and criticality based on MyDataCompliance questionnaire. The different types of data (e.g. with different classification or constraints or security needs) are described in MyDataCompliance, in order to evaluate the security risk for each sub-system. Please visit below link to know more about <https://www-2f95-p.epaas.eu.airbus.corp/folios/14099/compliance>

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5BAD		Configuration Airline Cockpit Version 1	
Alfabet ID:		APP-53513	
Classification : Airbus corporate Airbus AMBER Export control No restriction National Security No restriction Personal data Personal data			
Compliance : Status : assessed GR OK : 5 out of 18			
Questionnaire Progression		93/93 - 100% completed	

- The information managed by the System falls under the following categories:
 - Customer data (Airlines Data)
 - Personal data (users ex. email_id)
 - Competition data
 - Business Sensitive Data
- The information managed by the System does not fall under the following categories:
 - Export control data
 - Sensitive personal data
 - Private data

2.3.4. Extended Enterprise and Compliance

The application is planned to be exposed to airlines based in High risk countries. It will adhere to the security compliance requirements mentioned above. Additionally the HRC related compliance requirements will be added later, as HRC availability is out of scope for the initial version of the application.

As the application is using Internet Publishing, the supplier is entitled at WAF level to access the application using an airbus account (Airbus OneLogin).

2.3.5. Non Functional Requirements

The list of requirements below is the list of non-functional requirements derived from the constraints listed in this section of the document:

Cyber Security:

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ID	Short Name	Description	Note
AD-ACC-001	Accessibility	The system must be accessible to end users over the Internet through a Web Browser (Recent version of Google Chrome, Mozilla Firefox, Microsoft Edge or Apple Safari).	
AD-SEC-001	Authentication	All users accessing the System must be authenticated using SSO (OneLogin with Airbus provided account or accounts provided by Customer SSO through Identity Federation).	Internal:LAM-ADAM ET4GPMG groups ,for external DMZ
AD-SEC-002	Anonymous Access	Anonymous access to the System, its features or its data must be strictly forbidden.	
AD-SEC-003	Authorizations	Access to the System's features and data must be made according to fine grained authorizations. This can be achieved using the Skywise Authorization Service, Foundry teams, PCP user profiles...	SAS Authorization
AD-SEC-004	Security logging	All authentication or authorization errors must be logged with user login information (no password), date and time at a "SECURITY" log level to support security monitoring.	
AD-SEC-005	Data encryption at rest	All data must be encrypted at rest. This can be achieved either: - By enabling data storage encryption mechanisms - By directly encrypting the data to be stored if the data storage does not provide integrated encryption mechanisms.	
AD-SEC-006	Data encryption in transit	Data in transit must be encrypted. This can be achieved by either: - Using HTTPS or IPSec tunneling over the Internet with Airbus provided certificates - Using middleware encryption mechanisms on the PCP network - Directly encrypting the data if the network or middleware do not support/provide integrated encryption mechanisms.	
AD-SEC-007	Sensitive configuration data	Any sensitive configuration data (passwords, token...) must be stored in a secure manner. This can be performed on PCP using AWS System Manager's Parameter Store with SecureString and a dedicated encryption key accessible to application components and people responsible for secret management only.	
AD-SEC-008	External Developments	External developments must be performed in compliance with Airbus A1015.	

GDPR Compliance:

ID	Short Name	Description	Note
AD-GDPR-001	Data Policy	The System shall have a dedicated page describing its data collection policy. This data policy shall be maintained up-to-date and versioned according to evolutions of data collection needs.	
AD-GDPR-002	Consent Collection	The System shall systematically ask and record consent to the current version of the Data Policy from any new user, or from any registered users whenever the Data Policy is updated.	

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AD-GDPR-003	System Access	Access to the System shall only be granted to an user if that user has consented to the latest version of the Data Policy.	
AD-GDPR-004	Data Export	The System shall allow exporting the data pertaining to an associated airline and recorded by the System.	

Annex 13 Compliance:

ID	Short Name	Description	Note
AD-A13-001	Annex 13	The System shall allow System administrators to block access to data it contains and that relates to an aircraft under ICAO Annex 13 investigation.	
AD-A13-002	Investigation Support	The System shall allow System administrators to provide access to data it contains and that relates to an aircraft under ICAO Annex 13 investigation to people in charge of the investigation.	
AD-A13-003	Data preservation	The System shall not delete any data it contains in relation to an aircraft under ICAO Annex 13 investigation unless authorized to do so.	

Usability, Robustness, Operations:

ID	Short Name	Description	Note
AD-USA-001	Response Time	Application response time can be below 4 seconds for common requests. This response time may be higher for some specific requests or for some longer requests, the request will be registered and the user will be informed later when the result will be available.	
AD-USA-002	Loading Indicator	When response time is above 4 seconds, the user shall have an indication that the application is processing data and invite him to wait.	
AD-ROB-001	Server side robustness	The server side components must handle errors in a graceful manner allowing applications to be resilient to errors and to resume nominal operations without need for an intervention.	
AD-ROB-002	Data processing errors	In case of error while processing data, the erroneous data must be moved out of the processing pipeline to allow operations to resume nominally. Erroneous data can be stored in a dedicated area (DLQ, S3 bucket...) to allow troubleshooting operations.	
AD-MAI-001	Server side maintainability	Server side components must log application activity and all errors to allow application monitoring and maintenance.	
AD-SDO-001	Infrastructure as Code	All infrastructure shall be configured using architecture as code.	
AD-SDO-002	Deployments	Resource creation and application component deployment shall be made using the company automation services (Jenkins)	
AD-SDO-003	Code	All application code must be stored in the company code repository (Github) in skywise-maintenanceandmaterials organization	
AD-OPE-001	Production Access	No access to production accounts shall be required for nominal application operations.	

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2.4 System Architecture

This section presents the architecture as a whole and describes the rationale that led to the design of that architecture (choices of technology, COTS, architecture patterns...).

2.4.1. Architecture Drivers

Security

- Compliance to Airbus security rules.
- Compliance to Enterprise Architecture Principles
- Compliance to PCP rules and recommendations (Internet facing)
- The application is made available for the airline maintenance engineers as per the standard Airbus WAF configuration.

Data Compliance

- Contractual Compliance
 - The system will host data from different customers(Airlines), the System must therefore ensure that data confidentiality is maintained between customers and with Airbus according to contractual requirements.
 - Moreover, part of the data will be flowing through Foundry, and must therefore comply with Foundry contractual requirements and restrictions.
- Annex 13 Compliance
 - The System must be compliant with Airbus requirements regarding ICAO Annex 13 - Aircraft Accident & Incident Investigation in order to properly manage access to data in case of an incident involving an aircraft subscribed to the System.
- GDPR Compliance
 - The System shall be compliant with GDPR requirements and allow exporting and deleting a user's private data.
- Sustainable
 - The architecture needs to be **sustainable** which uses only optimum resources to compute and serve to the customer.

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2.4.2 Architecture Design Rationale

2.4.2.1. Platform

As discussed, the system shall be hosted across Skywise Foundry and Airbus AWS PCP platforms. The foundry platform hosts all the source dataset and code repositories to perform transformation of the dataset using pyspark.

The Airbus AWS PCP platform hosts the frontend and respective services exposed as application program interfaces to the frontend. It also hosts services to synchronise data with the Skywise Foundry platform using the walter exports.

Internet Facing: One of the major requirements/constraints is to have the application available/accessible on the internet. This is needed as the customers for this application are the Airlines & MROs who need to access this application on the internet. This has led to the selection of PCP as a platform for development, and to the use of WIPS blueprint setup by the PCP Team (Web Interface Publishing Serverless blueprint).

Federated Authentication Solution: The application uses Airbus OneLogin as authentication means. The users for the application must be registered in Airbus OneLogin and must be declared in the required AD groups on LAN-ADAM. This is essential and mandatory as it is a prerequisite for any application to be hosted on the internet from the PCP team.

Authorization via SAS: The application will re-use SAS for ICAO code segregation. If further fine-grained authorization is required for the upcoming Epics, SAS capabilities will be re-used. This is done to avoid the effort of creating a new Authorization service within Skywise and enhance re-usability.

Foundry data replication into AWS RDS: The data of the application will be fed by upstream data sources. The volume of data will be significant. There will be analysis performed on the data by performing multiple joins across all the data tables. The business requirement is to show the analysis data in realtime..To satisfy the above requirements, the relevant data will be initially cleaned and transformed in Foundry. Then the resultant data will be fed to AWS PCP s3 bucket using walter exports and then ingested into RDS using the global walter export project deployed on AWS PCP allowing real time analysis.

Serverless Components Priority: The application uses a serverless approach - the application uses AWS lambda (on demand computing) to serve the application -. This reduces the cost of operation and enhances maintainability. Other serverless components such as SNS, SQS, DynamoDB, Cloudfront, ApiGateway, Cognito, etc will be used as well.

Separate service for File Export : - XLS and CSV exports will be used across the application for multiple functionalities. A separate, loosely-coupled export service will be

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designed for the same. The same service can be used for other M&M and Skywise products going forward.

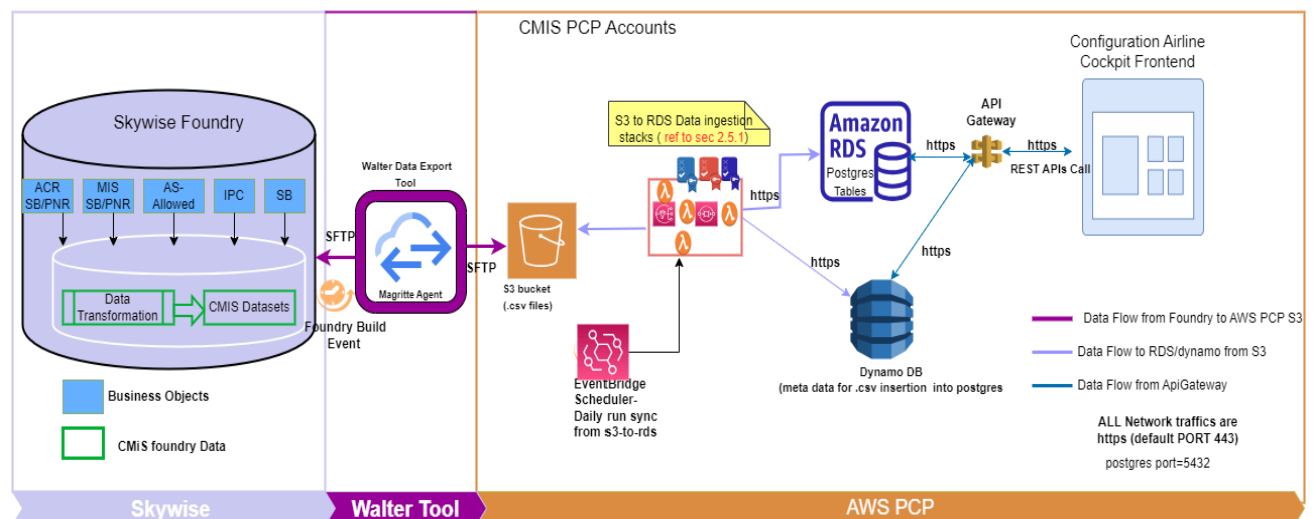
Infrastructure as code: The objective of infrastructure as code is to ensure reproducibility of the creation and configuration of the IT environment of the System, as well as to easily allow System's scalability.

Single Page Application: The Airbus AWS PCP platform hosts the frontend and respective services exposed as application program interfaces to the frontend.

2.4.2.2. Data

The data for the application gets ingested into the skywise foundry from Canibal AWS FLI and other data ingestion systems. The system uses data that is ingested in skywise foundry and performs all the necessary transformations as part of the foundry pipelines using pyspark. The transformed data consumable by the user interface is then synchronised with Airbus AWS PCP. The data is assembled in RDS aurora clusters, DynamoDB and S3 buckets based on the requirement.

Data Flow diagram



NOTE :

S3 to RDS Data ingestion (for more details ref to [sec 2.5.1](#))

Unnecessary data definition : Data which is no longer required for application to keep.

- S3 unused cvs :** all the csv which are exported from foundry to s3 bucket will be removed after 6 days , retention policy of 6 days.
- S3 exports file** (pdf, csv,xls) generated by application : wil be removed after 24hrs by scheduler.

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3. Aurora Postgres there is no unnecessary data which needs to be cleaned.

CMiS tables:

Below are transformed dataset corresponding based on business logic, for each postgres table which serve to the application presentation layer.

Data model link [[sb browse](#)] and [[parts browse](#)].

SB Browse Tables		SB PartsTables	
Foundry Datasets	Postgres Table	Foundry Datasets	Postgres Table
cmis_browser_sb_data_csv	cmis_browser_sb_data	aircraft_deatil_part	cmis_aircraft_detailart
cmis_sb_detail_csv	sb_details	cmis_acr_fin_pnr_map_csv	cmis_acr_fin_pnr_map
cmis_mp_detail_csv	mp_detail	cmis_acr_msn_fin_map_csv	cmis_acr_msn_fin_map
cmis_mod_detail_csv	mod_details	cmis_acr_msn_pnr_map_csv	cmis_acr_msn_pnr_map
cmis_browse_aircraft_detail_csv	aircraft_details	cmis_acr_part_browser_csv	cmis_acr_part_browser
cmis_mod_msn_map_csv	mod_msn_map	cmis_acr_part_description_csv	cmis_acr_part_desc
mod_sb_map_csv	mod_sb_map	cmis_mis_fin_pnr_map_csv	cmis_mis_fin_pnr_map
cmis_sb_msn_map_csv	sb_msn_map	cmis_mis_msn_fin_map_csv	cmis_mis_msn_fin_map
		cmis_mis_msn_pnr_map_csv	cmis_mis_msn_pnr_map
		cmis_mis_part_browser_csv	cmis_mis_part_browser

For more details abouts foundry Build schedule for [SB Browser](#) and [SB Parts](#) datasets.

List input datasets which cmis consume as upstream in foundry from

Business Objects Catalog :

Business Objects	Description
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Estimated in-service (SB)	Aircraft Configuration In Changes : All the data relative to the aircraft configuration in terms of changes: list of changes recommended or required (SB, MPM, MOD), changes incorporation status, dependencies between changes (SB/MPM). Format compliant with airbus Business Ontology model. Documentation is here: http://goto/ConfInChangesEstimatedInService Foundry Link https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.35663c6d-320c-49ec-aa2e-58d203ade059/master
Allowable_ConfInPart	List of parts allowed to be installed at a position for a given Aircraft Instance and a given Company. BO object : http://goto/ConfigurationInPartsAllowable Foundry Dataset link https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.be1ffd47-1cb1-47c9-b113-8f53d2ba4134/master
configuration_in_parts	Foundry dataset link https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.85195d48-80fc-43a0-9826-b867840d0bfb/master
[Line Maintenance - Airbus only - CDS] Theoretical Aircraft conf (Parts)	Program wise datasets : https://core.skywise.com/workspace/compass/view/ri.compass.main.folder.47984da3-b020-44d7-bade-9b1c6be61cde
Aircraft Configuration in Parts (most_probable_hbase)	Merge of Aircraft configuration data from the Operator's information system and the Airbus Theoretical configuration in parts. https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.06117c5a-f150-4a88-aa7f-41c6a3b021ce/master
[Full lifecycle] aircraft (aircraft)	The aircraft is the part of Aircraft Table full lifecycle, which provides the current state view. The Aircraft Table full lifecycle view provides aircraft table data for all civil aircraft, at each state of their lifecycle (pre and post-delivery). Format compliant with One Skywise Ontology. See documentation at: http://goto/ACTableFullLifecycle Foundry link https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.a9e649ad-e744-4ffb-a1fe-63e89f4abe89/master

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SB_Generic_Dependencies	SB Dependency : Gives information on possible dependencies for the accomplishment of the Service Bulletin: - may state any Service Bulletins which should not be accomplished prior to the accomplishment of the Service Bulletin - may require the previous accomplishment of one or more Service Bulletins Format compliant with Airbus Business Ontology model. See documentation: http://goto/TechdataEngineeringSBmetadata. Scope restrictions defined in: http://goto/TechdataEngineeringSBmetadataScope Foundry Link https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.30c20af5-0fd2-4e91-b083-4b04b9bdb4c8/master
FIN and CB	https://core.skywise.com/workspace/compass/view/ri.compass.main.folder.be5de438-3996-4ba4-b0fc-fd5237540eb5
Equipment Definition	https://core.skywise.com/workspace/data-integration/dataset/preview/ri.foundry.main.dataset.65933f97-5f7b-4718-b0ba-2e2300a3a357/master
Theoretical Aircraft Conf (Parts)	https://core.skywise.com/workspace/compass/view/ri.compass.main.folder.7b9ce4af-ee6b-42bf-a488-cecf2057a623

2.4.2.3. Security

The foundry platform leverages skywise group based secure access to all the dataset and code repositories used in the project. The data synchronisation to Airbus AWS PCP happens with the help of a secure system user credentials delegated for CMIS application. The transformed data to be displayed to the user interface shall only be extracted and synchronised with the AWS PCP platform.

The PCP platform also leverages Skywise Authorization Services and Cognito User Pool Federated with IAM Airbus One Login for Authentication. All the data stored in the AWS platform are encrypted at rest using a AWS KMS key for the application. The data is transmitted only through secured HTTPS connections. All the environment configuration parameters shall be stored in AWS parameter Store.

The web application shall be protected using WAF and imperva as per Airbus PCP standard guidelines.

2.4.2.4. Services

The system has two types of services, the northbound services that are exposed to the frontend via a secure API Gateway and the southbound services, that are typically

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triggered with the help of SNS or SQS, or Foundry Walter Data Export used to synchronize data between AWS PCP and Skywise Foundry platforms. The services shall be developed using AWS Lambda having nodejs / python as their runtimes. The architecture remains serverless as there are no dedicated server instances running round the clock to support the services. The pros and cons of the serverless architecture are mentioned below.

Pros:

1. Built in Scalability
2. Reduced Cost
3. Maintenance of Servers are managed by cloud service providers

Cons:

1. Multitenancy
2. Performance
3. Debugging and Testing

Mitigations:

In order to overcome security issues due to multi-tenancy, the lambda functions are supposed to perform only one specific task and they are supposed to be given just enough permissions to perform only to perform their designated tasks.

The performance benchmark of AWS lambda function clearly recommends using python / nodejs runtimes for best performance. The usage of API gateway caches and other caching mechanisms shall be encouraged. The lambda functions are recommended to be configured to use 256MB of memory and a 30 second timeout only, just enough to perform the only task it is supposed to perform. The memory and timeout shall be increased for specific lambda functions if required after discussion with tech leads and architects.

For the purpose of debugging, keeping the lambda code size is extremely important, as this helps the aws console to load the code and helps the developers debug effectively. All the lambda functions need to have exhaustive Unit Test cases that can be tested in the local machine before deployment to reduce the impact of testing limitations of serverless functions.

2.4.2.5. User Interface

The user interface of the application shall be developed using react javascript library. The user interface shall be developed according to the UX guidelines using airbus react design system. The frontend code shall be deployed in AWS S3 bucket and made available in edge location using cloudfront. The REST APIs are invoked using AWS Amplify library with authentication provided by AWS Cognito.

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2.4.3 Threat Modelling

A high level view of the system with the shared responsibility threat modelling is shown below.

For the purpose of threat modelling, let us employ the STRIDE threat model to discover the threats in the above system.

STRIDE is an acronym for the types of following threats:

Spoofing — a user or program pretends to be another

We have cognito authentication with JWT token in order to access the APIs in a secured manner. The validity of the JWT token is set to 1 hour and gets automatically refreshed. The token is set in the client browser using HTTPS for exchange to avoid token sniffing.

Ref [2.5.1.4 Authentication flow](#)

Tampering — attackers modify components or code

The code deployment permissions in AWS is restricted to specific devops ROLE assumed by Jenkins. The user roles do not have access permissions for manual code deployment other than the development account. The code runs through a series of checks before deployment in any AWS account. The code merged to the deployment branches only via pull requests with mandatory approvals. The developers need to secure their git access tokens and periodically renew the same.

The foundry repository write access is managed by using foundry skywise access controls. The build checks on foundry ensures that the new code deployment is complete only when the build check passes. The access is designated to teams and developers for code repositories.

Repudiation — threat events are not logged or monitored

The following logs and monitoring are to be configured in the system

1. Cloudfront Access Logs
2. API Gateway Access Logs
3. The logs from lambda functions integrated with SPLUNK
4. The foundry health checks integrated with SPLUNK
5. Imperva monitors application website traffic

Information disclosure — data is leaked or exposed

The data at rest in AWS systems are encrypted with a AWS KMS key. The foundry datasets are secured by palantir. To avoid any data leaks during transit, the data exchange between skywise foundry and Airbus AWS PCP only happens via HTTPs.

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Denial of service (DoS) — services or components are overloaded with traffic to prevent legitimate use

Imperva WAF monitors the web application and mitigates bot attacks and volumetric DDoS attacks.

Elevation of Privilege — attackers grant themselves additional privileges to gain greater control over a system

The JWT token is used to retrieve the user signature of the request and the data is validated with cognito and SAS before allowing elevated access to the users. The assignment of IAM users to the groups with elevated privileges are managed with the help of Airbus recommended tools.

The airbus one login owns the responsibility of user group associations for authentication and SAS owns responsibility for user authorization.

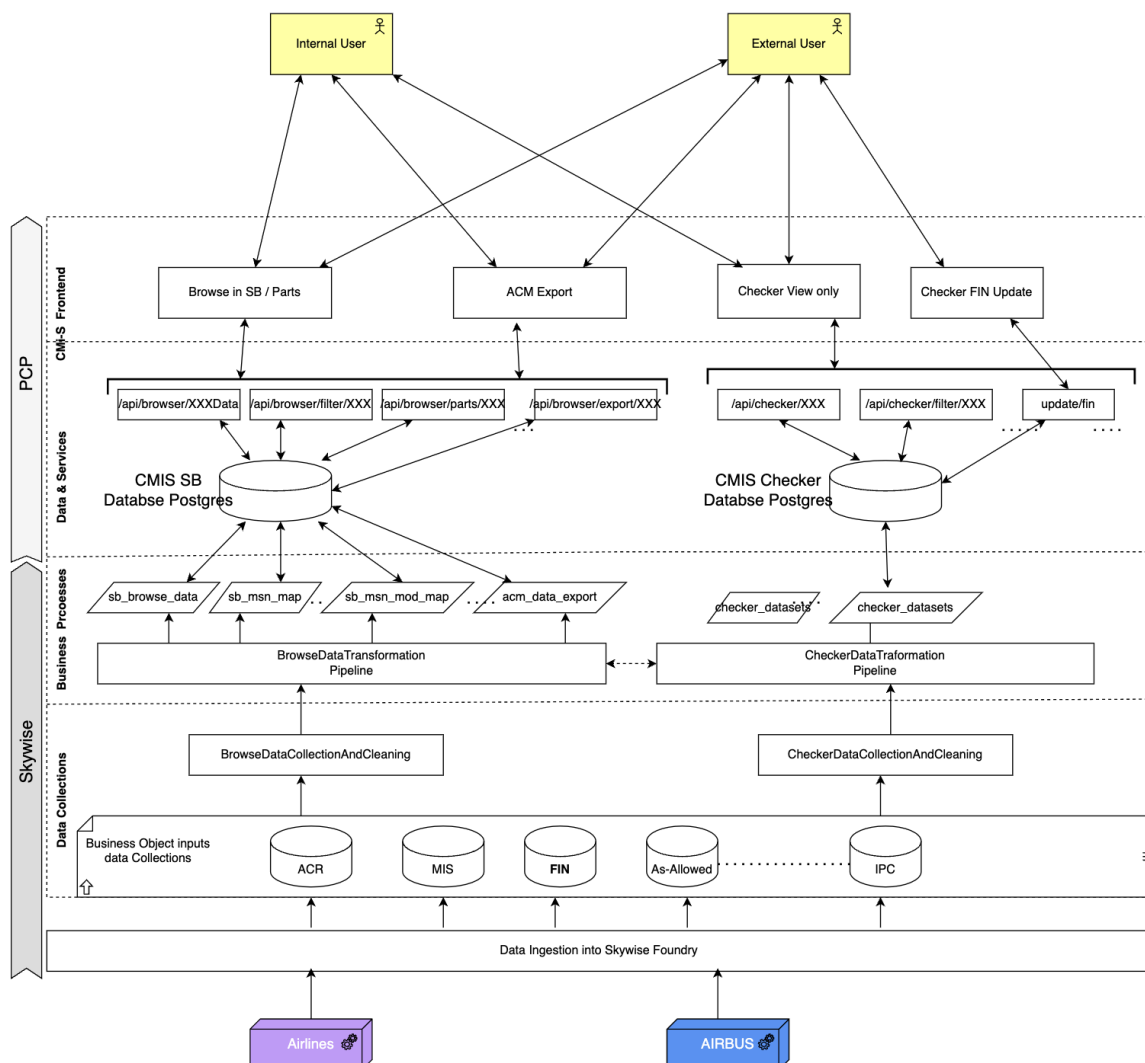
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2.4.4. Functional Application Mapping

*** Intentionally left blank, refer to 2.4.5 ***

2.4.5. Application View

The application view describes a more detailed view of the application layer with the general structure of the architecture without delving into a too-detailed description of each component and to description of the internal behaviour of the system, i.e. as it realises the expected services. This view identifies.



The application architecture follows closely the functional architecture.

Data Collections and Data processing:

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- **Business Object inputs data Collections** : The data for the application gets ingested into the skywise foundry from Canibal AWS FLI and other data ingestion systems. And publish as business objects in skywise business object catalog.
- **BrowseDataCollectionAndCleaning**: The system collects/copies the data into the system's foundry workspace/folder and performs all the necessary data cleaning transformations activities(cleaning , required columns selection,filtering,etc). Datasets output of this pipeline will be used further in browser data pipeline and checker data pipeline as mentioned below.
- **BrowseDataTransformation Pipeline**: This pipeline processes all the business functionality(data view for ACR,MIS,PNR,FIN etc) which is required by the Browse module to view the refined data at frontend.
This pipeline will transformed data and generate the datasets corresponding to each of functionality required to serve the frontend
- **CheckerDataTraformation Pipeline**: This pipeline processes all the business functionality (like computing the comparison between ACR and MIS vs As-Allowed configuration, etc) which is required by the checker module to view the refined data at frontend.This pipeline will transformed data and generate the datasets corresponding to each of functionality required to serve the frontend.

Each of datasets which produced by BrowseDataTransformation and CheckerDataTraformation Pipeline will be synchronized to AWS PCP rds Auora postgres database, corresponding to to each of datasets in foundry there will a table in RDS
Please refer to the dataflow topic above.

Services

- browse APIs :
- Checker APIs :
- ACM Export APIs:

To serve the frontend functionality we have an api for each of the functionality provided in the frontend.

For API details please [click here](#)

Frontend : frontend functionality can only serve services provided by the backend API services as mentioned above.

2.4.6 Architecture Deployment

The application is hosted on AWS PCP infrastructure and described as below:

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- Data locations:

#	Data	Location
1	Browse Data	Skywise Foundry,AWS S3, RDS Aurora Clusters/postgres, AWS PCP ,eu-west-1,Dynamo
2	Checker Data	Skywise Foundry,AWS S3,RDS Aurora Clusters/postgres, AWS PCP ,eu-west-1,Dynamo
3	ACM Export	Skywise Foundry,AWS S3,RDS Aurora Clusters/postgres, AWS PCP ,eu-west-1,Dynamo

The main services and transformations of the system are described below

#	Services and Transformations	Location
1	Browse services	AWS PCP Lambda eu-west-1
2	Checker services	AWS PCP Lambda eu-west-1
3	ACM Export	AWS PCP Lambda eu-west-1

2.5. System Components Architecture

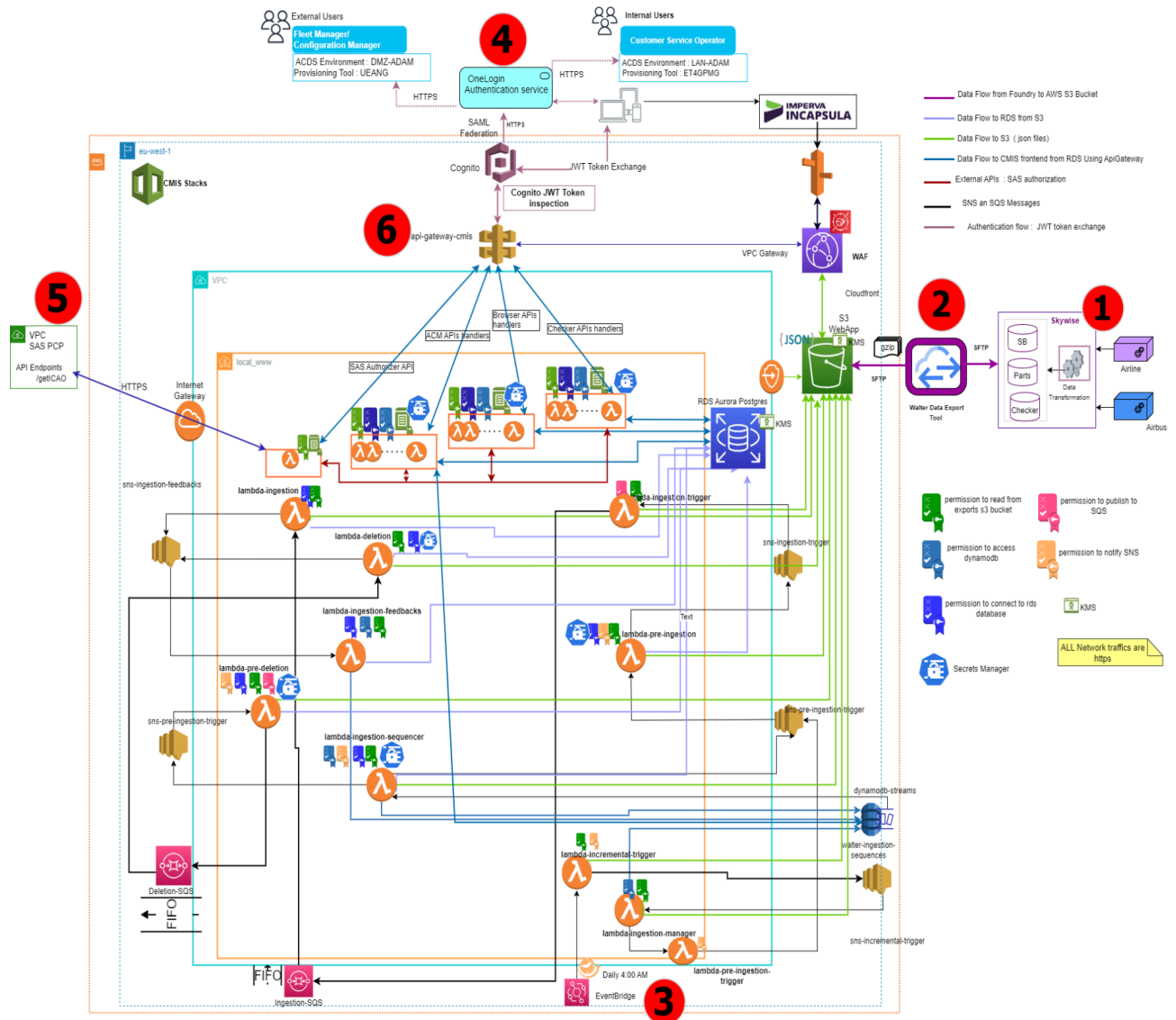
The following chapters present the System components' breakdown and details.

2.5.1. Architecture Breakdown Diagram

The Architecture breakdown diagram is an exhaustive, hierarchical tree structure of components that make up a product, arranged in a whole-part relationship. It is defined according to the system architecture decomposition.

Each component identified in the architecture breakdown is described in a specific following diagram.

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2.5.1.1 Skywise Transformation pipelines:

1

Skywise Transformation pipelines: Data pipelines are core of the application which transform

As per business logic defined by business owners. To develop the cmis data pipeline it happens in basically two steps.

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- 1) **Data collection and Data cleaning:** it will collect(copy) the data from Business Objects Catalog and perform data cleaning activities, basically preparing the data for the next pipeline. Based on the cmis functionality these components (**BrowseDataCollectionAndCleaning** and **BrowseDataCollectionAndCleaning**) are representing the same.
- 2) **Data processing:** as per business logic provided, based on that these components (BrowseDataTransformationPipeline, and CheckerDataTransformation Pipeline) are representing the data transformation and generate the datasets corresponding to each of functionality required to serve to the frontend.

For more details abouts foundry Build schedule for [SB Browser](#) and [SB Parts](#) datasets.

2.5.1.2 Walter Tool Data Export:



Walter Tool Data Export: after Data transformation we need to export the datasets to AWS PCP, for that we are using walter ,within the Walter Data ingestion tool there are multiple data export flows created[\[link\]](#). (As per application need for each table in Aurora postgres correspond to a dataset in skywise foundry).As soon as a dataset gets built in skywise foundry with new data(.csv format) walter flow will export the data to AWS PCP account S3 folder path (which is configured in walter export flow).

2.5.1.3 S3 to RDS Data ingestion stacks:



S3 to RDS Data ingestion stacks : Used to ingest the data to RDS. Once the data(csv files) is pushed to S3 by Walter Tool. A AWS EventBridge scheduler scans the s3 export bucket if new csv is added to the bucket then ingest it to respective rds tables.

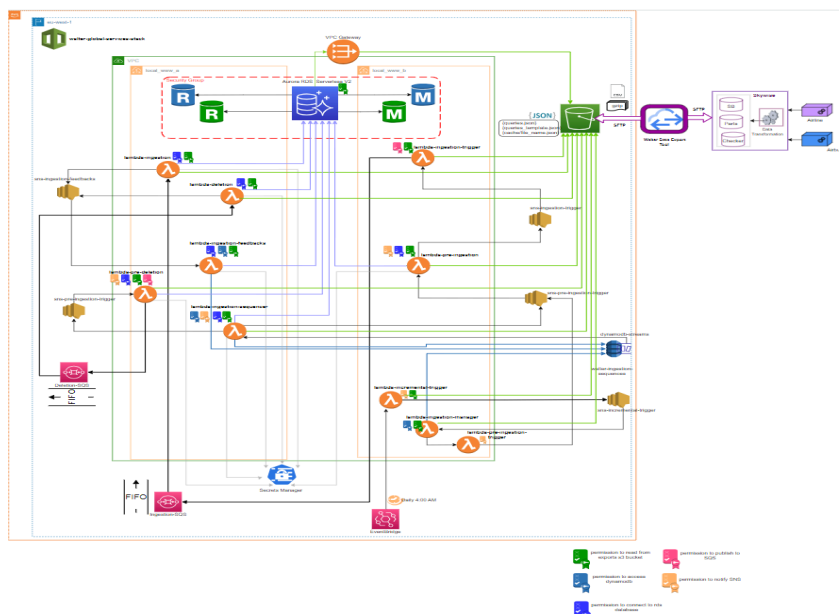
Lambda function used for data ingestions;

Function name	Description
lambda-5bad-walter-data-post-ingestion-pro	post ingestion use to validate the data after ingestion
lambda-5bad-walter-data-incremental-trigger-pro	for incremental data load trigger
lambda-5bad-walter-data-pre-deletion-pro	prepares deletion queries
lambda-5bad-walter-data-ingestion-trigger-pro	initiate RDS to copy data from S3 to table

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stack-5bad-walter-data-im-BucketNotifications Handl-ROOmPVumLP4o	AWS CloudFormation handler for "Custom::S3BucketNotifications" resources (@aws-cdk/aws-s3)
lambda-5bad-walter-data-ingestion-feedbacks-pro	feedback after every ingestion / deletion jobs are completed
lambda-5bad-walter-upload-object-pro	used for uploading json files in s3
lambda-5bad-update-bucket-trigger-pro	updates the meta informations of uploaded files - gzip encoding
lambda-5bad-walter-data-ingestion-manager-pro	managing all the sequences and callbacks, dynamo stream
lambda-5bad-walter-pre-ingestion-trigger-pro	initiation of data ingestion, notifies SNS, lambda which is getting trigger by EventBridge scheduler
lambda-5bad-walter-pre-ingestion-pro	this function used execute ddl sql script (creating tables etc)
lambda-5bad-walter-data-ingestion-pro	function used to ingest the data to rds
lambda-5bad-walter-data-ingestion-sequencer-pro	managing sequences for each ingestion pro
lambda-5bad-walter-data-deletion-pro	deleting the data from rds tables

The below simplified diagram explains the data ingestion from s3 to rds Tables



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2.5.1.4 Authentication flow:

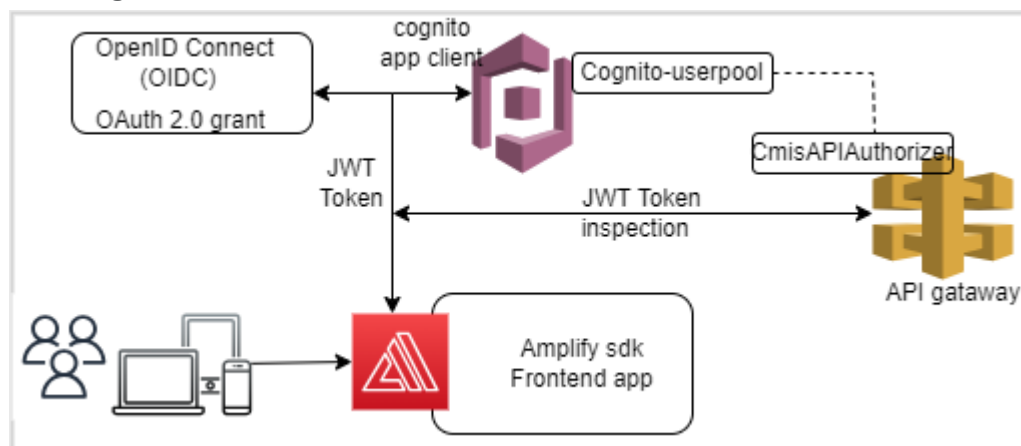


Authentication flow :

AWS Amplify lib has been used with Amazon Cognito as the main authentication provider. The authentication is managed by OneLogin (through Identity Federation).

We have cognito authentication with JWT token in order to access the APIs in a secured manner. The validity of the JWT token is set to 1 hour and gets automatically refreshed. The token is set in the client browser using HTTPS for exchange to avoid token sniffing.

Simplified diagram for authentication and JWT token flow



Cognito app client configuration

Identity providers	OAuth grant types	OpenID Connect scopes	Authentication flows
Cognito user pool directory	Authorization code grant	Email,openid,aws.cognito.signin.user.admin, phone	ALLOW_REFRESH_TOKEN_AUTH, ALLOW_CUSTOM_AUTH,
airbus-onelogin			

JWT Token details

Refresh token expiration	Access token expiration	ID token expiration
1 day	1hr	1hr

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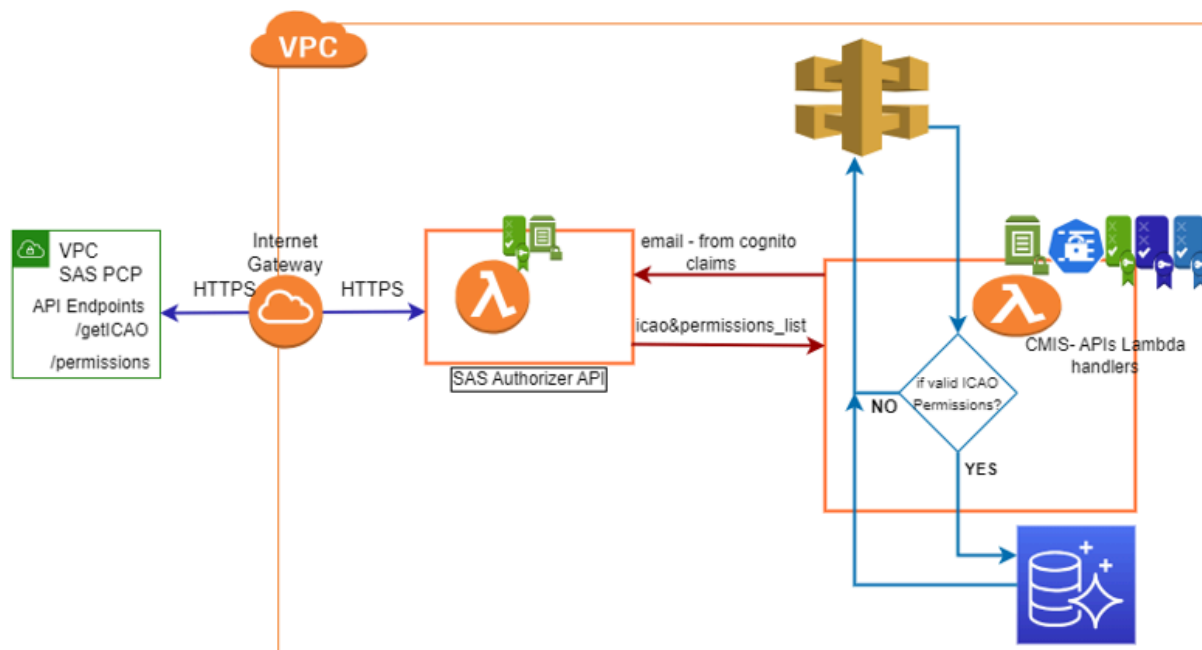
All the APIs exposed by the configuration cockpit(CMIS) application are secured by cognito authorizer. Only users with valid airbus onelogin and added to Et4GPMG group will have a valid JWT to access the APIs.

Ref: ([2.6.2.4.6.2. ET4GPMG Groups](#))

2.5.1.5 SAS Authorization overview:

- 5 **SAS Authorization:** As Soon as users get authenticated by onelogin, cmis app will make SAS Authorization service call (REST API) to get user authorization details (user association with ICAO and cmis Application permission which configured), if Authorization failed ui component will not load.All CMIS APIs will be guarded by SAS authorization (permission will get validated for every APIs call) . Only if the user belongs to that ICAO and has valid permission(ref [2.6.2.4.6.1. Account Management Process](#)) then only the api will return the successful request.otherwise “unauthorised access”. And failure attempts will be logged.

Simplified diagram for Authorization flow :



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2.5.1.6 CMIS APIs:

6 CMIS APIs: (**Browse APIs**, **Checker APIs**, and **ACM APIs**) : REST APIs exposed to the cmis UI components, based on task UI operation it will serve the UI functionality. Predominantly READ operations on the data. Each API handler (lambda function) has two sections. First section is SAS Authorization validation(pls refer to point 4) ,second section interact with AWS data persistence layers composed of dynamodb, aurora RDS clusters, S3 etc. For API details please [click here](#)

The application uses only HTTPS (**default PORT 443**) for the interactions with external systems

Method	API	Description(Short)	Ref Dataset(RDS)	Lambda Function
POST	/api/browser/parts/mis	Fetch part details from MIS including Installation date and operator code	cmis_browser_part_mis_data	lambda-5bad-cmis-part-mis-handler-pro
POST	/api/browser/parts/acr	Fetch part details from acr including hardware function identifier	cmis_browser_part_acr_data	lambda-5bad-cmis-part-acr-handler-pro
POST	/api/browser/parts/operator Msn	Fetch part msn and data source	cmis_browser_part_aircraft_detail	lambda-5bad-cmis-part-operator-msn-handler-pro
POST	/api/browser/parts/pnrMis	Fetch pnr details from mis including hardware function identifier	cmis_browser_part_pnr_mis	lambda-5bad-cmis-part-mis-handler-pro
POST	/api/browser/parts/pnrAcr	Fetch pnr details from acr including hardware function identifier	cmis_browser_part_pnr_acr	lambda-5bad-cmis-part-pnr-acr-handler-pro
POST	/api/browser/parts/finMis	Fetch fin details from mis including hardware function identifier	cmis_browser_part_fin_mis	lambda-5bad-cmis-part-fin-mis-handler-pro
POST	/api/browser/parts/finAcr	Fetch fin details from acr including hardware function identifier	cmis_browser_part_fin_acr	lambda-5bad-cmis-part-fin-acr-handler-pro
GET	/api/browser/filter/operator	Fetch distinct operators for internal users form Database	cmis_aircraft_details	lambda-5bad-cmis-filter-operator-pro

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GET	/api/browser/filter/actype?param=	Fetch distinct aircraft types of given operators for internal users form Database	cmis_aircraft_details	lambda-5bad-cmis-filter-actype-pro
GET	/api/browser/filter/actype?param=&aircraft_program_code=	Fetch distinct aircraft types of given operators and aircraft family for internal users form Database	cmis_aircraft_details	lambda-5bad-cmis-filter-actype-pro
GET	/api/browser/filter/acfamily?param=	Fetch distinct aircraft families of given operators for internal users form Database	cmis_aircraft_details	lambda-5bad-cmis-filter-acfamily-pro
POST	/api/browser/sbData	Fetch data from Database for Table view based on multiple filters from frontend.	cmis_browse_sb_data	lambda-5bad-cmis-SBD ataView-pro
POST	/api/browser/filter/msn	For Fetching distinct id_aircraft_registration of given operator and actype	cmis_aircraft_details	lambda-5bad-cmis-filter-msn-pro
POST	/api/browser/filter/sb	For Fetching distinct sb_reference of given id_aircraft_registration	aircraft_details, sb_msn_map	lambda-5bad-cmis-filter-sb-pro
POST	/api/browser/filter/mod	For Fetching distinct mod_reference of given id_aircraft_registration	aircraft_details, mod_aircraft_map	lambda-5bad-cmis-filter-mod-pro
GET	/api/browser/exportUrl	Fetch the pre-signed url to download the exported file	NA	lambda-5bad-cmis-export-file-pro
GET	/api/browser/filter/eStatus	Fetch download status of the exported file	NA	lambda-5bad-cmis-export-status-pro
POST	/api/browser/exportFile	Insert/update record in dynamodb table	cmis_browse_sb_data, cmis_browser_part_pnr_acr	lambda-5bad-cmis-export-save-file-pro
POST	/api/checker/fleet_detail	Fetch data from database based on given AC Type, id_aircraft_registration, operator and other filters	checker_fleet_details	lambda-5bad-cmis-fleet-detail-pro
POST	/api/checker/fleet	Fetch data from database based on given Operator, AC Type, MSN, and Disperency result.	checker_cmis_fleet	lambda-5bad-cmis-fleet-view-pro
GET	/api/user	For a authorised user SAS response is returned.	NA	lambda-5bad-cmis-auth orization-entry-pro
GET	/api/sas?email=	For a given authorised user SAS response is returned	NA	lambda-5bad-cmis-auth orization-pro

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2.5.2. External Interfaces

The following table lists the different components implemented by external systems with which the System will be interfaced:

Component	Source	Interface Details
Users Authentication	OneLogin/Cognito	The authentication is managed by OneLogin (through Identity Federation)
User authorizations	Skywise Authorization Service (SAS)	REST APIs over HTTPS. Refer to the section (2.5.1. Architecture Breakdown Diagram point 4)

2.5.3. Secure Development Lifecycle

The secure development lifecycle describes the security features of the SDLC process to design, develop and test application software components.

Security checks is managed with :

- Use of npm audit (and fix)
- GitHub (Security advisory support by Dependabot)
- Sonarqube Checks
- Imperva Monitoring

Testing Framework for E2E (in context of the MVP):

Frontend: Automated Robot framework for frontend flow testing.

APIs Testing : As of manual API testing has been going on **[Automation is planned]**

a test plan shall be defined to ensure the following aspects:

- Proper multi tenancy implementation - A user from an airline XYZ shall not be able to view and access data from another airline company ABC
- Proper base security implementation - Anonymous access to APIs shall be blocked
- Proper implementation of segregation between Internal and external users permissions.

Tests shall be performed systematically, either manually or automatically, after any new delivery and before any opening to end users.

DevOps toolchain will be set with :

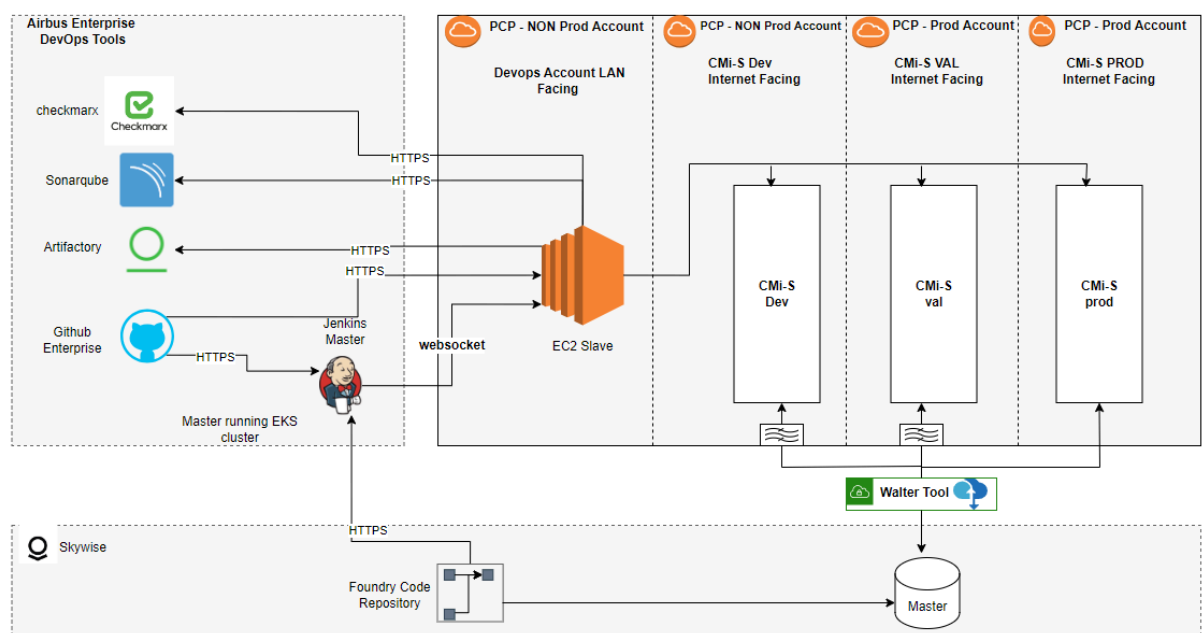
- GitHub Enterprise (Airbus service) for source code management for UI services, Sync services and frontend.
- Code Repository (Skywise foundry service) for source code management for transformation pipelines.

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- Artifactory (Airbus service) for artifact management (npm and pip packages) COTS download and build publishing.
- Jenkins (Airbus service) for CI and CD job orchestration.
- Sonarqube (Airbus service) for code quality analysis and build checks.
- Checkmarx (Airbus service) for SAST (Static Application Security Testing). [planned but not integrated as of now]

Jenkins calls GitHub and SonarQube APIs authenticated by tokens generated from a service account. PCP team has provisioned all PCP accounts with role-devops-jenkins-agent role. Each account must allow the DevOps account to assume this role in order to allow the deployment steps (through AWS CDK). Github enterprise is integrated with pull request checks in Jenkins server, so that any changes to the main and develop branches will undergo static checks, unit tests before being deployed.

An AWS PCP LAN facing non PROD account is setup with a EC2 jenkins slave to perform the build orchestrations.



An AWS PCP internet facing non PROD account is set up with two deployment stages dev and int. The code from the develop branch is deployed to the dev and int stages. The developer can make deployments from local machines to any stage in the AWS non PROD account. The developer can connect a user interface running in a local development server to any REST API service in the non PROD account.

An AWS PCP internet facing PROD account is set up with two deployment stages, val and pro. The code from the main branch is deployed to the val and pro stages. Only automated deployments are allowed to make changes in the AWS PCP PROD accounts.

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Developers are granted read only access to the resources in the AWS PCP PROD account for debugging purposes.

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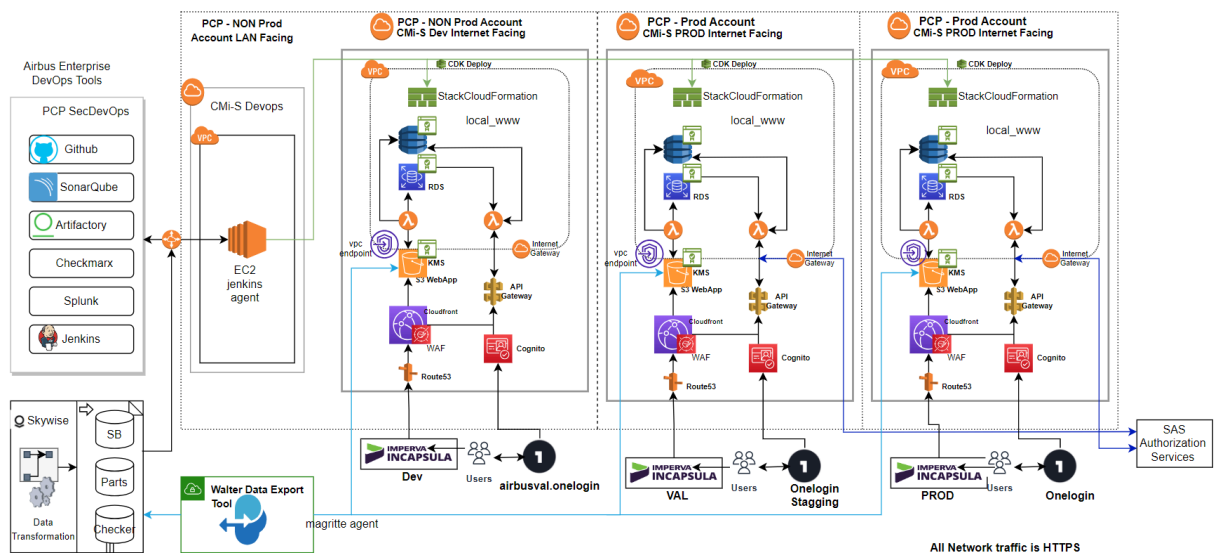
2.6. System Infrastructure

This section describes the Infrastructure layer for the global system and how this layer meets security concerns.

2.6.1. Infrastructure Overview

2.6.1.1 Infrastructure Logical View

The infrastructure logical view identifies and describes all the hardware servers and clients to be used by the system in the operational environment. The logical view shows the main infrastructure components with the service they implement and their relationships within the system. External interfaces with external applications are also described.



2.6.1.1 Infrastructure Physical View

The physical characteristics of the logical infrastructure components are described below.

	Component	Standard Description	Remarks
ARD-EIC-010	VPC	eu-west-1	
ARD-EIC-011	Subnet	/26 2 subnets in each AZ	
ARD-EIC-012	Interface Endpoint Service		
ARD-EIC-013	Gateway Endpoint	256MB memory, 30s Timeout	
ARD-EIC-014	VPC Endpoint	Endpoints has been configured to import s3 file to RDS aurora	
ARD-EIC-015	S3	No Public Access , No Versioning,	

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		KMS Encrypted	
ARD-EIC-015	DynamoDB	On Demand, 0 Replication regions, KMS Encrypted	Indices can be created based on use cases
ARD-EIC-016	Lambda	256MB memory, 30s Timeout	
ARD-EIC-017	RDS	Aurora Postgres PostgresEngineVersion.VER_15_2, InstanceClass.R6G, InstanceSize.XLARGE2 storageType:AURORA_IOPT1	
ARD-EIC-018	CloudFront	Price Class 100, WAF enabled, custom ACM certificate, Access Log Enabled, TLSv1.2_2021	
ARD-EIC-019	API Gateway	Edge optimised, Caching Enabled, 10000 request per second throttling with 5000 request bursts. Web ACL and Logging Enabled	Standard REST API endpoint definitions with GET, POST, PUT and DELETE methods
ARD-EIC-020	Cognito	Federated with Airbus OneLogin, SAML based	
ARD-EIC-021	WAF		
ARD-EIC-022	Walter		
ARD-EIC-023	Foundry Dataset		

2.6.2. Security Overview

2.6.2.1 Backup and Restore Strategy

Disaster recovery (DR) refers to preparing for and recovering from a disaster. A disaster can be any negative event that can impact the Airbus or its customers' business continuity, like hardware or software failures, a network or a power outage, human error or some other significant event.

The two important industry terms for preparing for a disaster are **Recovery Time Objective (RTO)** and **Recovery Point Objective (RPO)**.

While CMIS is running on AWS PCP technology, when reacting to a disaster, it is essential to either quickly commission resources or to orchestrate the failover to already running resources.

Regions: The CMIS Platform is hosted on AWS PCP in eu-west-1 region and uses as much as possible all Availability zones allocated for Ireland Country as long as Amazon delivers services on those regions (*3 in total with some services limitation*).

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Storage: Making sure that the data persisted in the infrastructure is not lost during an outage is essential for the CMIS application. The main service used for storing the data is AWS RDS which uses multiple Availability Zones available within the region.

The SLA for AWS RDS is guaranteed¹ by Amazon to make each Multi-AZ DB Instance and each Multi-AZ DB Cluster available with a Monthly Uptime Percentage as shown in the table below during any monthly billing cycle (the "Multi-AZ DB Instance and Multi-AZ DB Cluster SLA")

	Resources	Monthly Uptime Percentage	Service Credit Percentage
ARD-DRP-010	Multi AZ DB Instance & Cluster SLA ²	Less than 99.5% but equal to or greater than 99.0%	10%
ARD-DRP-011	Multi AZ DB Instance & Cluster SLA ²	Less than 99.0% but equal to or greater than 95.0%	25%
ARD-DRP-012	Multi AZ DB Instance & Cluster SLA ²	Less than 95.0%	100%
ARD-DRP-013	S3 Standard SLA ³	Less than 99.9% but greater than or equal to 99.0%	10%
ARD-DRP-014	S3 Standard SLA ³	Less than 99.0% but greater than or equal to 95.0%	25%
ARD-DRP-015	S3 Standard SLA ³	Less than 95.0%	100%
ARD-DRP-016	DynamoDB Standard & Global Tables SLA ⁴	Less than 99.99% but equal to or greater than 99.0%	10%
ARD-DRP-017	DynamoDB Standard & Global Tables SLA ⁴	Less than 99.0% but equal to or greater than 95.0%	25%
ARD-DRP-018	DynamoDB Standard & Global Tables SLA ⁴	Less than 95.0%	100%

Service Credits

Service Credits are calculated as a percentage of the charges paid by you for the Multi-AZ DB Instances or Multi-AZ DB Clusters that did not meet the Multi-AZ DB Instance and

¹ <https://aws.amazon.com/rds/sla/>

² <https://aws.amazon.com/rds/aurora/sla/>

³ <https://aws.amazon.com/s3/sla/>

⁴ <https://aws.amazon.com/dynamodb/sla/>

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Multi-AZ DB Cluster SLA, as applicable; or for the Single-DB Instances that did not meet the Single-DB Instance SLA in the affected AWS region for the monthly billing cycle in which the SLA, as applicable, was not met.

Networking: when an outage occurs, the connectivity between end customers and the CMIS application is impacted and service cannot be delivered. This service continuity is handled by Amazon Cloud Infrastructure accordingly with its SLA² between the range of 99.95% and 99.99%. As Cloud Solution Provider, Amazon is responsible for delivering its services in the Ireland region and Airbus Operational team implements adequate measures to ensure minimum outage on Skywise Platform. However, as the global design of CMIS application relies mostly on AWS PCP, very few isolated/dedicated elements need to be replicated or backedup.

More details on backup and restoration can be found [here](#).

2.6.2.2 Cryptography Management

Cryptography in Skywise Foundry is managed by Airbus Security within Foundry as per Palantir and Airbus best practices.

2.6.2.2.1 Encryption at rest:

All the data storage (S3, DynamoDB, DocumentDB) and middleware (SNS, SQS) at rest is encrypted by a single region symmetric AWS Custom KMS key.

The web application source code and assets are stored in a separate S3 bucket that has data encryption with standard server side encryption AES256 algorithm.

2.6.2.2.2 Encryption in transit:

The data is also always secured in transit by using HTTPS with TLS 1.2. We have multiple connections to the storage :

Magritte Agent - Skywise data transit: Magritte agent communicates data to the skywise platform using HTTPS.

Encryption of data in transit is AWS Custom SSL certificate (The RSA 2048 Asymmetric encrypted data exchange)

SSL Certificates

- AWS Certificates - Amazon Certificate manager (ref 2.6.2.4.3 Certificates:)
- SSL Certificates - Airbus PKI

All the data flows, protocol followed and the ports are mentioned in the Network Policies section. All data flows in the application follow HTTPS protocol. The application uses TLS V1.2 (minimum).

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2.6.2.2.3 Key Management

Since the application is hosted on AWS PCP platform, we manage the keys using AWS key management services (KMS). All keys created for the application are customer managed keys.

Key	Key Spec	used by resources	Managed by	Key rotation
kms-5bad-default-key-global	SYMMETRIC_DEFAULT	APi handlers Lambda Functions	arn:aws:iam::0595702XX:role/role-user-administrator", arn:aws:iam::0595702XX:role/airbus-service-role/devops/role-devops-jenkins-agent", arn:aws:iam::0595702XX:root"	Automatically rotate this KMS key every year
		RDS		
		Dynamo DB		
		S3		
		SNS		
		SQS		
kms-s3-encryption-walter-key	SYMMETRIC_DEFAULT	S3		
		SNS		
		SQS		
		RDS		
		Data ingestion Lambda Functions		

2.6.2.3. Auditing and Logging

The following information must be logged and audited.

For Unauthorised access.

As the system hosts multiple Airlines data, below events(failure of Authorization) will be logged with “**SECURITY**” Tag

- Authenticated airline user trying to access data from competitors data

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- System components must log any tentative made by a user to access to data out of what it is authorised to access to.

API Logs:

The API gateway access logs have the following format

```
{
  "requestId": "$context.requestId",
  "userAgent": "$context.identity.userAgent",
  "sourceIp": "$context.identity.sourceIp",
  "requestTime": "$context.requestTime",
  "httpMethod": "$context.httpMethod",
  "path": "$context.path", "status":
  "$context.status",
  "responseLength": "$context.responseLength",
  "resourcePath": "$context.resourcePath"
}
```

cloud front : The cloud front access logs have the following fields

```
Fields: date time x-edge-location sc-bytes c-ip cs-method cs(Host)
cs-uri-stem sc-status cs(Referer) cs(User-Agent) cs-uri-query
cs(Cookie) x-edge-result-type x-edge-request-id x-host-header
cs-protocol cs-bytes time-taken x-forwarded-for ssl-protocol
ssl-cipher x-edge-response-result-type cs-protocol-version
fle-status fle-encrypted-fields c-port time-to-first-byte
x-edge-detailed-result-type sc-content-type sc-content-len
sc-range-start sc-range-end
```

2.6.2.4 Security Maintenance

2.6.2.4.1 OS and antivirus

The System components will principally be using managed services (AWS, DevOps...). Therefore, in that regard, no update or patching of the OS is expected. Yet, the Project is to take into consideration any security notice issued by the providers of the used managed services.

2.6.2.4.2 Source code

The Project is expected to immediately correct any major or critical security issue discovered either by Checkmarx security audits.

Similarly, the project is expected to immediately take into account any security advisory issued by the GitHub Dependabot plugin.

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2.6.2.4.3 Certificates

Certificates used by the System are managed by Airbus and are out of the scope of the Project.

Domain Name	Region	Issued& Managed By	Renewal eligibility	Public key info	Signature algorithm	Associated resources
cmis-5bad-configairlinecockpit-pub.aws.airbus.cloud	us-east-1	Amazon	yearly - Managed by AWS	RSA 2048	SHA-256 with RSA	arn:aws:cloudfront::059570240523:distribution/E3OW3LEUC3LEWE

2.6.2.4.4 Keys

Encryption keys are managed using the AWS KMS and shall not require any maintenance. Nevertheless a key might have to be renewed in case it is foreseen as being compromised. API keys might have an expiration date and must be managed accordingly.

The security maintenance section describes the security operating model of the application (e.g. Software Configuration Management, application maintenance processes and account management).

For more details ref to [2.6.2.2.3 Key Management](#)

2.6.2.4.5 COTS

All COTS vulnerabilities are scanned (OWASP scan) by github **dependabot** and sonarqube. Reports can be seen in sonarQube for each repos

1. [AIRBUS/5bad-cmis-fe](#)
2. [AIRBUS/5bad-cmis-fe-cdk](#)
3. [AIRBUS/5bad-cmis-be](#)
4. [AIRBUS/5bad-cmis-be-cdk](#)

For the transformation pipelines, standard dev, int, val and master branching strategy is followed in code repositories. The changes to all these branches are allowed only with the help of pull requests on respective code repositories in skywise foundry.

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2.6.2.4.6 Application Maintenance Process

The application maintenance and support activities are carried out using the standard 3 level organisation end to end Support model.

2.6.2.4.6.1. Account Management Process

Permissions associated with functionality.

UserType	Role Name	SAS Permissions Applied in PROD	Application functionality Mapping
External(Air lines)	Fleet Manager/Configuration Manager	CMIS_BROWSER_RW_AL, CMIS_CHECKER_RW_AL, CMIS_ACM_RW_AL	Browse functionality Checkers functionality ACM functionality
Internal	Customer Service Operator	CMIS_BROWSER_R_ABUS CMIS_ACM_R_ABUS	Browse functionality ACM Browse functionality

Functionality associated with User Type

UserType	Role Name	Browse Functionality	ACM functionality	Checkers functionality
External	Fleet Manager/Configuration Manager	Yes(readOnly)	Yes(readOnly)	Yes (read and write both)
Internal	Customer Service Operator	Yes (ReadOnly)	Yes (ReadOnly)	No

Where :

No - Do not have any permission

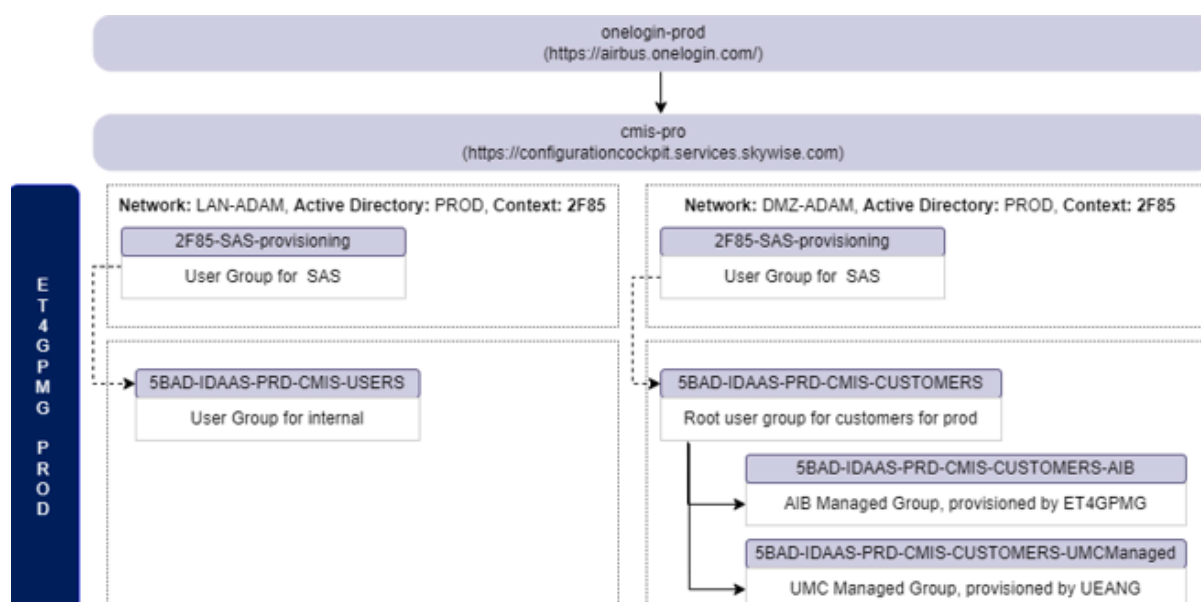
2.6.2.4.6.2. ET4GPMG Groups

To access the application user need be part of ET4GPMG groups mentioned as below based on internal and external users.

ET4GPMG Group Names	Description	Remarks
---------------------	-------------	---------

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5BAD-IDAAS-***-CMIS-USERS	Root group for internal users, synchronized with one-login	Internal LAN users can be added here using ET4GPMG.
5BAD-IDAAS-***-CMIS-CUSTOMERS	Root group for external users, synchronized with one-login	DO NOT add users in this group directly
5BAD-IDAAS-***-CMIS-CUSTOMERS-AIB	Nested in 5BAD-IDAAS-***-CMIS-CUSTOMERS	Provisioned by ET4GPMG, Users can be added here using ET4GPMG. For managing access to developers and internal users in DMZ. AIB (ICAO for Airbus)
5BAD-IDAAS-***-CMIS-CUSTOMERS-UMCManaged	Nested in 5BAD-IDAAS-***-CMIS-CUSTOMERS	Provisioned by UEANG , to manage customers
ET4GPMG-2F85-SAS-provisioning	SAS provisioning group present in both LAN-ADAM and DMZ-ADAM network	Root groups are to be nested inside this group to automatically add users added in the group to be part of SAS users. Requires super context creation between 5BAD and 2F85 (SAS)



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2.6.2.5 Endpoint Protection

Backend REST APIs: APIs must ensure any call is issued by an authenticated user, and data is accessed according to explicit authorizations (no access to data by default).

Airbus AWS PCP standard endpoint protection pattern is applied with the help of AWS WAF and imperva.

2.6.2.6 xMan

The development tools used do not require xMan.

2.6.2.7 Annex 13

Standard procedure in case of Annex 13 will rely on:

- SAS for the System front-end
 - Whenever an aircraft is put under Annex 13 by the authorities, that aircraft must be also indicated as under Annex 13 in SAS. This will prevent SAS from listing this aircraft as accessible to end-users.
 - The fact that the System correctly applies authorizations provided by SAS must be the subject of dedicated testing
- Markings in Foundry
 - Whenever an aircraft is put under Annex 13 by the authorities, that aircraft must be also indicated as under Annex 13 in Foundry. This will trigger a rebuild of all datasets marked as Annex 13.
 - The fact that the System's datasets have the correct Annex 13 marking must be audited periodically.
- **PCP system** : As of now Annex13 will be handled on support request basis by Dev Team, not automated **[Planned in future Version of development]**

2.6.2.8 Pentest click [here](#) for access pentest report

2.6.2.9 Security Risk Assessment report : [click here](#)

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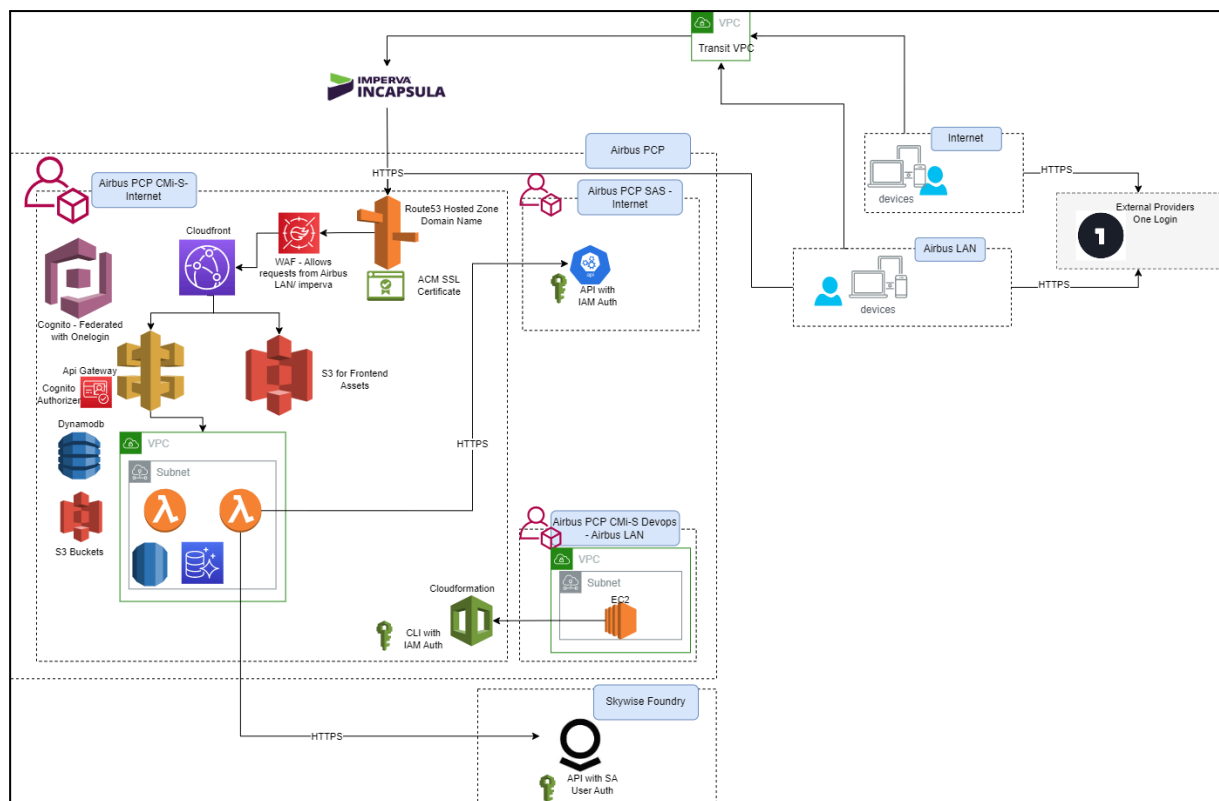
2.6.3 Network Overview

2.6.3.1 External Network

The external network overview with interactions with the following external systems

1. Airbus One Login
2. Airbus PCP SAS
3. Skywise Foundry

The application is accessible from Airbus LAN network and open internet by using the imperva URL. The interactions with the external systems AirNavX and SRS are confined to client level url navigation and there are no other system interactions. The network view of external applications is beyond the scope of this document.



2.6.3.2 Internal Network

The structure of the internal network supporting the deployment of the system and the locations of the main hardware nodes connected to it. The AWS PCP resources are deployed in the eu-west-1 region. The resources are deployed in a VPC and there are 2 subnets configured in each availability zone inside the VPC.

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VPC	/25 vpc-ew1-5bad-configairlinecockpit-prod
Subnets	subnet-ew1-5bad-configairlinecockpit-prod-local_www-b subnet-ew1-5bad-configairlinecockpit-prod-local_www-a
Security Group	secg-ew1-platform-egressonly

2.6.3.3 Network Security

The network security section describes the expected network security topology for this project such as logical connection requirements, segregation capability of components and network protections solutions. The following security mechanisms shall be implemented to ensure security of the data at transit. None of the resources are allowed direct access from the public internet.

- Cognito Federated with Airbus One Login:** All the APIs exposed by the configuration cockpit(CMIS) application application are secured by cognito authorizer. Ref [2.5.1.4 Authentication flow](#)
- Only users with valid airbus onelogin and added to configuration airline cockpit (CMIS) ET4GPMG group will have a valid JWT to access the APIs.
- System User Token for foundry APIs:** For the APIs to foundry, a system user token is generated and used as authorization bearer JWT.
- Skywise Authorization Services:** AWS IAM based authorization is enabled for the SAS APIs to retrieve the user's fine grained authorization permissions.
- IAM Based Role for Cloudformation Deployment:** IAM based permission is granted to the EC2 machine used as Jenkins slave to perform deployments in CMIS PCP accounts.
- AWS Roles and Policies for Lambda:** For the lambda functions specific roles and permissions are created by the cloud formation infrastructure as code stacks, these lambda functions are given just enough permissions to access resources. These lambda instances are executed in specific VPCs.]

2.6.3.4 Network Flow Matrix

Flow	From	To	Protocol	Port	Authentication	Authorization	Account Type	Use Case
1	Client Browser	Imperva Incapsula	HTTPS	443	One Login	Coarse grained by OneLogin + AD Group; Fine Grained by SAS	Personal Account	Client connection to application

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2	Imperva Incapsula	Cloudfront	HTTPS	443	Using IP whitelisting on the Cloudfront ACL		-	-
3	Cloudfront	Frontend s3	-	-	Cloudfront OAI		Cloudfront OAI	Connect cloudfront to static UI files
4	Cloudfront	API Gateway	HTTPS	443	Cognito	Fine grained by application	PCP Account	Connect UI to services
5	API Gateway	SAS	HTTPS	443	X-api key	Application code	PCP Account	Get Fine grained access info of the user
6	API Gateway	Lambda			Policies			Route queries
7	API Gateway	Cognito						Check Authentication token header
8	Cloudwatch	Lambda run						Every 12 hours detect repetitive actions
9	Lambda Run	DynamoDB	HTTPS	443	Access Policies			Store user preferences
10	Lambda Run	RDS		5432	Access Policies			Read and write data
11	Lambda	S3	HTTPS		Access Policies			To store xls and csv files

2.6.4 IAM

This section describes for each type of users (e.g. non-privileged users, privileged users and technical admin users) and technical accounts (e.g. system accounts, service accounts, etc.) identification, authentication and authorization controls in place

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2.6.4.1 Anonymous Privileged Users

No anonymous access to the System is authorised. All users must be authenticated through OneLogin to access the System.

2.6.4.2 Non Privileged Users

For internal users : have "ET4GPMG" as a Provisioning Tool.

Identification: ACDS Environment :LAN-ADAM

ET4GPMG : ET4GPMG-5BAD-IDAAS-PRD-CMIS-USERS

Authentication: Airbus Onelogin

User Type with Permissions configured in SAS Application based on Airline contract definition.

UserType	Role Name	SAS Permissions Applied in PROD	Application functionality Mapping
External(Airlines)	Fleet Manager	CMIS_BROWSER_RW_AL, CMIS_CHECKER_RW_AL, CMIS_ACM_RW_AL	Browse functionality Checkers functionality ACM functionality
Internal	Customer Service Operator	CMIS_BROWSER_R_ABUS CMIS_CHECKER_R_ABUS, CMIS_ACM_R_ABUS	Browse functionality read only Checkers functionality read only ACM read only

2.6.4.3 Privileged Users:

For External users: have "UEANG" as Provisioning Tool, and follow the naming convention:

Identification: ACDS Environment : DMZ-ADAM ,

ET4GPMG : UEANG-5BAD-IDAAS-PRD-CMIS-CUSTOMERS-UMCManaged

Authentication: Airbus Onelogin

Authorization: AWS Cognito and SAS

Only External Users can change the status of FIN and add the comment for a particular FIN in checker module's FIN details based on successful authentication and authorization by SAS.

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2.6.4.4 Bastion System Administrators

None Identified

2.6.4.5 M2M Accounts

sa-5bad-devops: Account used for devops

- The git credentials token for Jenkins (ssh/ https)
- The artifactory credentials for using airbus jfrog artifactory
- The credentials for sonarqube project analysis
- The above credentials are stored in SSM (of devops PCP account) as secure string
- Password expire default 180 days :
- The rest of the password and storing SSM is manual.

SA-5BAD-CMIS-API :for foundry code checkout by jenkins jobs.

This SA user has editor access to the foundry workspace.

- Password expire default 180 days :
- The rest of the password and storing SSM is manual.

WALTER Data Ingestion System : This application has access to the Datasets for exporting to AWS S3.

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2.7. Architecture Assessment

The purpose of this section is to assess the global architecture in regards with critical operational scenarios and global simplification of the IM landscape.

2.7.1. Architecture Quality Goals

A quality attribute that the system needs to achieve. These goals often have long term character in contrast to (short term) project goals.

2.7.2 Performance and Scalability

The performance and scalability aspects of the architecture determines the ability of the system to predictably execute within the mandated performance profile and to handle increased processing volumes.

2.7.2.1 Scenarios

Expected performance scenarios stated by business:

1. As a Fleet Manager, I should be able to view the browser in parts/SB and based on the datasource and filter applicability.
2. As a Fleet Manager, I should be able to view the checker view on the datasource and filter applicability

2.7.2.2. Performance Assessment

ID	Description
ARD-PAS-010	Overall maximum number of user expected on the application: 500-1000
ARD-PAS-011	Expected amount of data to be managed:1TB
ARD-PAS-012	Expected percentage of data modified each day by end users: NO Modification by app end users
ARD-PAS-013	Expected percentage of data modified monthly by upstream sources: 20%
ARD-PAS-014	Expected max concurrent user connections: ~100
ARD-PAS-015	Any page rendering should not take more than 5 seconds
ARD-PAS-016	Any export operation should not take more than 5 minutes

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2.7.2.3. Scalability Assessment

For the AWS Lambda functions, the default regional concurrency quota starts at 1,000 instances. For an initial burst of traffic, the cumulative concurrency can reach between 500 and 3000 instances per minute.

The AWS DynamoDB is set to **on-demand**, as application traffic is difficult to predict and control and the application workload has large spikes of short duration. Amazon Aurora clusters are set to **Provisioned instance db.r6g.2xlarge**, Instance classes powered by AWS Graviton2 processors. These instance classes are ideal for running memory-intensive workloads. I/O optimised - enabled via console for improved price performance in case of I/O intensive apps.

By enabling caching for AWS API Gateway. It caches responses from an endpoint for a specified time-to-live (TTL) period. API Gateway then responds to the request by looking up the endpoint response from the cache instead of making a request to lambda services. The default TTL value for API caching is **300 seconds**. Only the GET methods have caching enabled by default. This helps to ensure the safety and availability of API.

For the transformation functions in the skywise foundry, output tasks will write to individual partitions sized at no more than **128MB**. This enables them to be sufficiently large without bleeding into unacceptable job run-times. Parquet files of this size avoid per-file overhead associated with metadata reading and file listing. As the partition sizes grow, begin to see that the trade-off between write and read time is that writing larger files will lose the advantages of parallelism but largely worth the read performance gains.

2.7.3. Availability and Reliability

The availability and reliability aspects of the architecture ensures a highly available application along with the process of recovering from failures such as data loss, major downtime. The Failure Mode Analysis builds resiliency into an application early in the design stage. It helps to identify the types of failures the application might experience, the potential effects of each, and possible recovery strategies.

2.7.3.1 Scenarios

1. Failure of SAS Authorization requests
2. Failure of data synchronization with skywise foundry (via : Walter export S3 => RDS)
3. Failure of data transformation pipelines to build the data

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2.7.3.2 Availability and Reliability Assessment

The standard characteristics of PaaS solutions are sufficient to fulfil the target 99% Availability and Reliability requirements for Configuration Airline Cockpit.

ID	Description
ARD-ARA-010	Failure of SAS auth requests, all the application services will be down.
ARD-ARA-011	In case of failure of sync with skywise foundry, the application shall continue to work with the available data (last sync data)
ARD-ARA-012	In case of failure of transformation pipelines, the application shall continue to work with the last built data

2.7.4. Cost Optimization

2.7.4.1. Scenarios

Expected cost optimization scenarios

- Pay only for the computing resources that you require and increase or decrease usage depending on business requirements, not by using elaborate forecasting.
- Continuously monitor cost and usage and review existing architectural decisions to ensure they continue to be the most cost effective.
- Foundry Data transformation pipeline : Compute only for new lines or new data

2.7.4.2. Cost Optimization Assessment

The cost optimization assessment explains how the architecture will be a cost effective solution while serving the right service level.

ID	Description
ARD-COA-010	AWS PaaS solutions will be used to minimise infrastructure cost
ARD-COA-011	API GW and Cloudfront caching will be used to minimise lambda cost
ARD-COA-012	DynamoDB single table design will be attempted to minimise the RCU and WCU
ARD-COA-013	Lambda functions are set to minimum compute per second by configuring 256MB of memory and 30s as default timeout. The lambda that requires more computation are reviewed and configured.
ARD-COA-014	For large sized dependencies, lambda layers are used to reduce the size of the lambda

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ARD-COA-015	The datasets for different data analysis are build separately , instead of taking it from the same base table thereby reducing the runtime computations in lambda
ARD-COA-017	The minimised final computed data required by the application for building the data models are only synchronised to AWS from skywise foundry
ARD-COA-018	Foundry data transformation happened only for newly added records (incrementally using hash function to identified change in existing or new)

2.7.5. Operational Excellence

The operational excellence of the application focuses on methodologies to continuously improve the operation of the application.

2.7.5.1 Scenarios

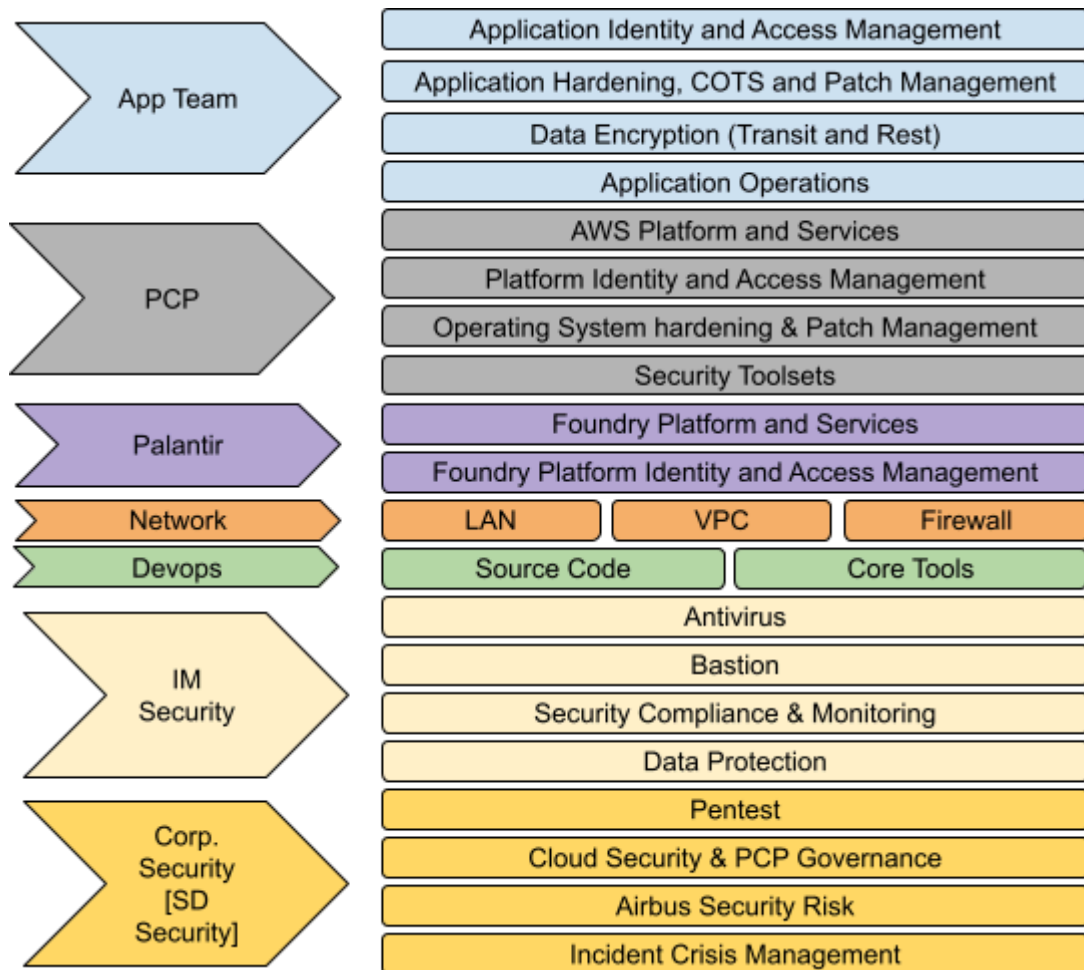
The methodologies typically involve people, process and technology.

- Refinement, Planning and Analysis
- Detailed design and use of industrialised patterns
- Clean coding and Unit testing
- Validation and Test Automation Strategy
- Automatic on demand deployment with feature toggles
- Real Time Monitoring and Support Activities

2.7.5.2. Operational Excellence Assessment

The operational excellence assessment explains how the architecture will support operational excellence objectives described by the different identified requirements / scenarios. The following schema is a global view of the shared responsibility model that applies to the Configuration Airline Cockpit application.

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The operational excellence assessment explains how the architecture will support operational excellence objectives described by the different identified requirements / scenarios.

ID	Description
ARD-OEA-010	AWS PaaS solutions will be used to minimise infrastructure operational efforts
ARD-OEA-011	Lambda autoscaling will be used to manage load variation
ARD-OEA-012	Existing authentication and authorization providers will be used
ARD-OEA-013	Splunk will be used by L2 team for real-time monitoring and alerts
ARD-OEA-014	Usage of AWS PCP blueprints for Web Internet Publishing Serverless
ARD-OEA-015	Usage of AWS PCP ASGARD (Jenkins agents JNLP) built with the CDK blueprint
ARD-OEA-016	Devops pipeline with build breakers configured for sonarqube based on code coverage, maintainability, reliability, duplication and security indices

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ARD-OEA-017	Pull Requests to be verified with quality checks and peer reviews before merge
ARD-OEA-018	Checkmarx1 build breakers for sec devops security by design
ARD-OEA-019	Automated test cases for E2E tests using selenium and robot framework. Automated API tests using robot framework.
ARD-OEA-020	Feature toggles for UI and Services for on demand deployments in production.

2.7.6. Complexity Reduction

The Application shall consist of a classic 3-tiers architecture: presentation layer (web browser), business layer (web server) and data layer (database servers). The architecture implements serverless microservices choreography with event sourcing. Each lambda function adheres to SOLID design principles. The permissions to read / update data and call services are restricted to the designated lambda function.

Choreography: Each microservice publishes events that may trigger other service(s).

Event Sourcing: Data handling driven by sequence of events.

Pros:

- Decentralised application with decoupled services
- Enforced cohesion with functionally focused services
- Parallel and fast processing
- Smaller and Cleaner Code

2.8. Components Catalogue

The following table identifies the software and hardware components that are planned to be used by the application (with relevant versions).

- On the client
- On the server

The **STAMP Standard Status** column indicates if the component is

- **Standard** (identified as standard component, i.e. managed by the software category owning product),
- **Specific** (not a standard component but agreed to be deployed for the specific use case and to be managed by the application owning product for the time being) or
- **Non standard** (not agreed to be deployed).

Platform type	Platform hardware	Software Component	Version	STAMP Standard Status
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SERVERS	AWS EC2 Jenkins Slave	Linux	ami-airbus-pcp -amazonlinux2 -release-20230 614-1023	Standard
CLIENTS	Ends user's PC	Windows	>10	Standard
		Mac		
		Linux		
		Chrome	>90	Standard

We have licence checkers configured in airbus enterprise sonarqube for the projects that displays the COTS component used and its corresponding licence. These are continuously updated for every build.

COTS used in project code repositories
cmis-frontend
cmis-frontend-cdk
cmis-be-cdk
cmis-services-cdk

2.8.1. Components Hardening

The components hardening describes system and software configuration hardening for Airbus solutions where hardening is not supposed to be already in place. Hardening implies secure configuration management but also endpoint protection solutions and integrity controls.

We have used serverless architecture in the application using lambdas and AWS cloud native services, we have mentioned the configuration of the EC2 in the Components Catalogue section. Hence the hardening session is empty

Hostname	Physical host	Type of environment	OS version	Network zone	Site location	Hardening Guides

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3. System Requirements and Traceability

3.1. Traceability to business requirements and other sources

This section is out of scope.

3.2. Impacted Work Products

None identified.